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Structure-based design of flavonoid inhibitors for MCL-1, an anti-apoptotic cancer target with no approved drug, how linker chemistry, aromatic surface area, and scaffold polarity govern recognition of a shallow, largely hydrophobic protein groove.

A controlled nine-compound series built on three flavone scaffolds, ranked by docking and then followed into 100 ns of all-atom dynamics. The design hypothesis was that hydrophobic burial alone could carry the binding. The simulation showed that it could, and also that it was not the whole story.

Insights: Structure-Based Drug Design, Molecular Docking, Structure-Activity Relationship, Molecular Dynamics, MM/GBSA Binding Free Energy, Ligand Efficiency.

Computational Design and Evaluation of Chr-CH_2-Naphthyl Derivative Targeting the MCL-1 BH_3-Binding Groove via Molecular Docking and Dynamics Simulations for Apoptosis-Based Cancer Therapy.

Pharmacological Research - Reports (Manuscript Under Review), Manuscript Number: PREREP-D-26-00437. Mahadesh Chandro Mondal, Md. Jabbir Ali, *et al.*

My contribution: Conceptualization, Visualization, Validation, Data curation, Investigation, Methodology, Analysis, Writing - original draft, Writing - review & editing, Resources, Software.

The design

We fixed the modification at the C7 hydroxyl of three flavones, Chrysin, Luteolin, and Wogonin, so we could vary linker chemistry and aromatic size one at a time rather than together. Nine derivatives, partially factorial: the aryl-size comparison within the ether-linked set, the linker comparison at the naphthyl terminus.

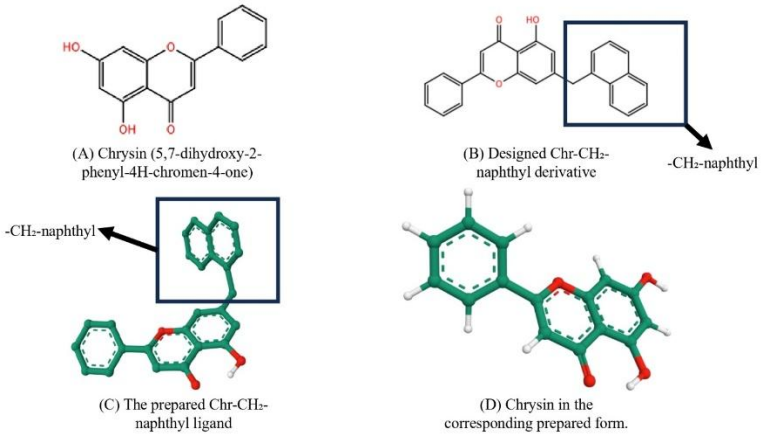
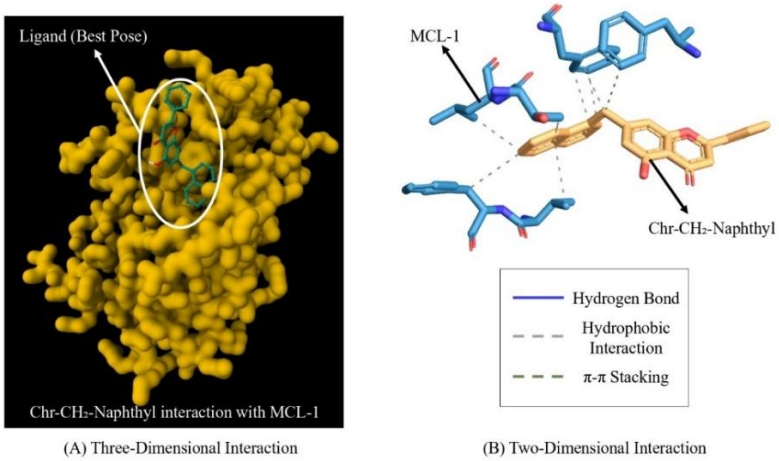


Figure 1: *(a) Chrysin (5,7-dihydroxy-2-phenyl-4H-chromen-4-one), SMILES O=c1cc(-c2ccccc2)oc2cc(O)cc(O)c12, (b) The designed derivative Chr-CH_2-Naphthyl, 5-hydroxy-7-[(naphthalen-1-yl)methyl]-2-phenyl-4H-chromen-4-one, SMILES O=c1cc(-c2ccccc2)oc2cc(Cc3cccc4ccccc34)cc(O)c12, (c) The prepared Chr-CH_2-Naphthyl ligand following ETKDGv3 embedding and MMFF94 geometry optimization, as submitted for docking. (d) Chrysin in the corresponding prepared form.*

Compound	Score (kcal·mol⁻¹)	Δ vs parent	H-bonds
Chrysin	−7.66	—	1
Chr-O-CH_2-Phenyl	−8.26	−0.60	1
Chr-O-CH_2-Naphthyl	−8.95	−1.29	0
Chr-CH_2-Naphthyl	**−9.44**	**−1.78**	**0**

Molecular Docking

Every derivative bound more strongly than its parent, and the two design variables separated cleanly because only one moved at a time. A naphthyl terminus in place of phenyl gained 0.70 to 1.03 kcal·mol⁻¹ across all three scaffolds; removing the C7 oxygen for a direct carbon linker gained a further 0.34 to 0.49, and the three carbon-linked compounds took the top three scores (-9.44, -9.35, -9.33) while forming no detectable hydrogen bonds at all. The parent ranking is the more instructive result: Luteolin, the most hydroxylated of the three, bound weakest at -6.88, while the least substituted Chrysin bound best at -7.66; added polar functionality pays a desolvation penalty at this groove without buying complementarity, the opposite of flavonoid behaviour at more polar targets. The reference inhibitor S-63845 scored -8.46 under the same protocol, though scores are not comparable across scaffolds of that size difference, and its 3.20 Å redocking RMSD limits the protocol to site identification and within-series ranking. The gain is also not free: ligand efficiency falls from roughly 0.40 kcal·mol⁻¹ per heavy atom for Chrysin to 0.33 for the lead.



(A) Three-Dimensional Interaction (B) Two-Dimensional Interaction

Figure 2: *Binding mode of Chr-CH_2-Naphthyl within the BH_3 groove of MCL-1 obtained by molecular docking. (a) Three-dimensional representation, with the receptor shown as a molecular surface and the ligand, in its top-ranked conformation, in stick representation within the binding site. (b) Interaction diagram of the same pose generated with PLIP, in which the ligand is shown in orange and the contacting residues in blue.*

100 ns Molecular Dynamics Simulation

The complex held throughout: mean C_α RMSD 1.32 ± 0.16 Å, radius of gyration varying by less than half a percent, seven of eight pocket residues fluctuating at or below the 0.80 Å protein mean, and a minimum contact distance of 2.96 ± 0.22 Å with no unbinding event. The number that matters is 52.3 ns. ARG263 was in contact from the start, but only through its aliphatic arm; the guanidinium sat 7 to 8 Å from the flavone carbonyl for the first fifty nanoseconds, then closed, and the hydrogen bond held at 100% occupancy across the final 45 ns. Docking had detected no hydrogen bonding in this pose, and the compound had been chosen precisely to test whether hydrophobic burial alone could carry high-affinity binding. In my view, this is a useful result rather than a favorable one: the hypothesis was partly wrong, and the way it was wrong is what matters; a rigid-receptor calculation cannot reach an interaction that requires 50 ns and mutual adjustment of both partners. MM/GBSA returned -11.19 ± 1.17 kcal·mol^{-1}, van der Waals at -14.80 ± 1.52 supplying roughly 70% of the gas-phase total, with electrostatics at -6.27 ± 0.91, larger than the static pose implied, and consistent with the late hydrogen bond. Conformational entropy was not estimated, so this is an effective enthalpy, comparable within protocol rather than an affinity prediction.

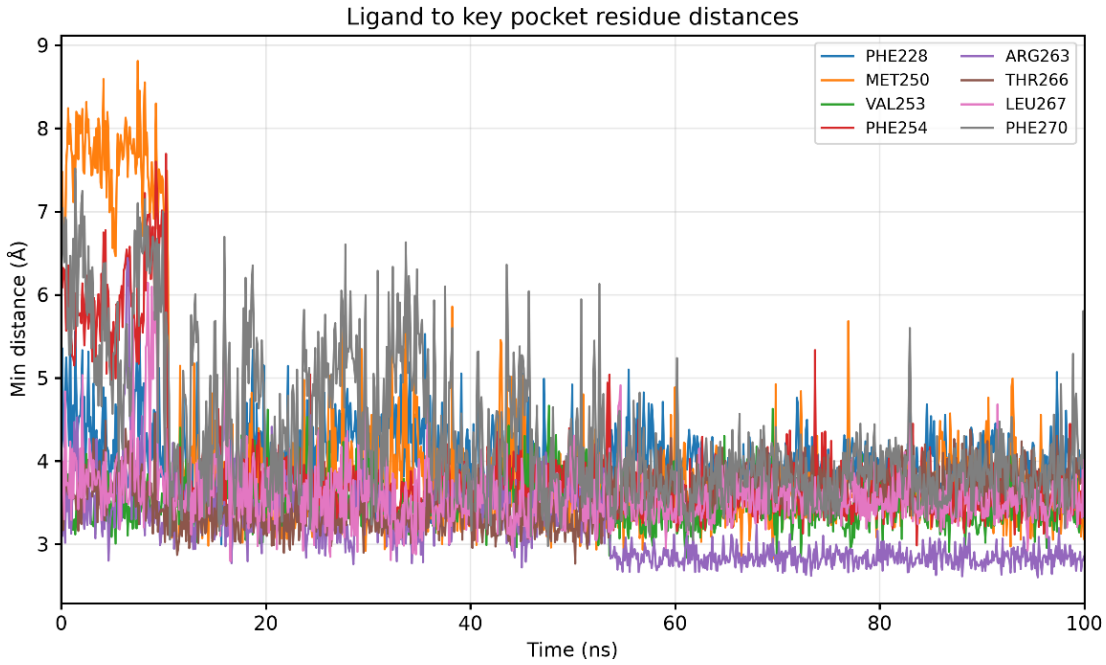


Figure 3: *Time evolution of the minimum distance between Chr-CH_2-Naphthyl and the eight pocket residues (PHE228, MET250, VAL253, PHE254, ARG263, THR266, LEU267 and PHE270). The hydrophobic residues MET250, PHE254, and PHE270 close in over the first 10-15 ns as the ligand relaxes from the rigid-receptor docked geometry, while ARG263 forms a hydrogen bond to the flavone C4 carbonyl at 52.3 ns, holding a mean separation of 2.84 Å with 100% occupancy for the remainder of the simulation.*

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